IoT TEMPERATURE AND GAS MONITORING USING ESP32

A MINI PROJECT REPORT

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ABSTRACT

Monitoring environmental temperature and the presence of combustible gases is important in homes, laboratories, and industrial spaces, where sudden changes in conditions can affect safety and comfort. This project presents the design and development of an IoT Temperature and Gas Monitoring system built around the ESP32 microcontroller. The primary objective is to develop a simple, affordable, and compact system that continuously measures environmental conditions and displays them locally for the user.

The proposed system uses a DHT22 sensor to measure temperature, while MQ-2 and MQ-5 gas sensors provide readings based on their response to particular gases and vapours. The ESP32 collects the sensor outputs, processes them, and compares the readings with predefined threshold values to determine the environmental status. The readings and the resulting status are then presented on an LCD display. The system is designed for local monitoring and operates without depending on an internet connection.

The project emphasizes the practical use of embedded systems, sensor interfacing, and real-time data display. Its simple design, low cost, and ease of assembly make it suitable for educational use and basic environmental monitoring applications. Because the ESP32 has built-in wireless connectivity, the system also provides a foundation for future enhancements such as wireless data transmission, mobile application integration, cloud-based storage, graphical data visualization, automatic data logging, and additional warning indicators.

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1. INTRODUCTION

1.1 OBJECTIVES

The primary objective of this project is to design and develop a simple and cost-effective system that monitors environmental temperature and gas levels using the ESP32 microcontroller and displays the readings locally on an LCD.

The specific objectives of the project are:

1. **Temperature Measurement:** To measure the environmental temperature accurately using the DHT22 sensor.

2. **Gas Monitoring:** To monitor the presence of gases and vapours using the MQ-2 and MQ-5 gas sensors.

3. **Data Processing:** To collect and process the sensor readings using the ESP32 microcontroller.

4. **Threshold Evaluation:** To compare the readings against predefined threshold values in order to determine the environmental status.

5. **Local Display:** To present the temperature and gas sensor readings clearly on an LCD display.

6. **Offline Operation:** To enable local monitoring without depending on an internet connection.

7. **Low Cost and Compact Design:** To build an affordable, compact, and easily assembled system suitable for educational and basic monitoring applications.

8. **Performance Evaluation:** To verify the working of each sensor and evaluate the monitoring performance under different environmental conditions.

9. **Future Enhancement:** To provide a foundation for wireless data transmission, mobile application integration, cloud storage, and data logging.

1.2 PROJECT OVERVIEW

Temperature and gas levels are important indicators of the condition of an environment. An unexpected rise in temperature or the presence of combustible gases and smoke can indicate unsafe conditions in residential, industrial, and laboratory settings. A simple and cost-effective monitoring system that allows these changes to be observed easily is therefore very useful.

- **System Development:** Design and implementation of an ESP32-based system for monitoring temperature and gas levels.

- **Environmental Sensing:** Continuous collection of temperature data from the DHT22 sensor and gas data from the MQ-2 and MQ-5 sensors.

- **Data Processing:** Processing of sensor readings by the ESP32 and comparison with predefined threshold values.

- **Local Display:** Presentation of the readings and the environmental status on the LCD.

- **System Integration:** Integration of the microcontroller, sensors, display, and power supply into a working prototype.

- **Experimental Validation:** Recording of readings under different environmental conditions to evaluate system performance.

- **Scope for Improvement:** Potential extension toward wireless connectivity, mobile applications, and cloud-based monitoring.

1.2.1 Role of the ESP32 and Sensors

The ESP32 acts as the central processing unit of the system. It reads the outputs of the sensors, processes the data, and sends the information to the LCD for display. The DHT22 sensor measures the environmental temperature, whereas the MQ-2 and MQ-5 sensors respond to particular gases and vapours present in the surrounding air. Together, these components form a compact monitoring unit that can be used in educational and basic environmental monitoring applications.

[Insert original photo of the complete project prototype here]

Figure 1: Complete prototype of the IoT Temperature and Gas Monitoring system

1.3 DIRECT BENEFITS

- Enables continuous monitoring of environmental temperature.
- Provides readings related to gases and vapours using the MQ-2 and MQ-5 sensors.
- Displays the readings instantly on a local LCD.
- Operates without an internet connection.
- Uses low-cost and easily available components.
- Simple design that is easy to assemble and use.
- Supports basic safety awareness by indicating changes in environmental conditions.

1.4 INDIRECT BENEFITS

- Enhances practical knowledge of embedded systems and sensor interfacing.
- Develops problem-solving and programming skills.
- Provides hands-on experience with the ESP32 and the Arduino development environment.
- Encourages innovation in IoT-based monitoring applications.
- Serves as a foundation for advanced systems with cloud connectivity and data analytics.
- Promotes interdisciplinary learning in electronics and programming.

1.5 ORGANIZATION OF THE REPORT

This report is organized into six chapters, each describing a specific aspect of the design, development, and implementation of the project.

Chapter 1 - Introduction: This chapter introduces the importance of monitoring environmental temperature and gas levels in residential, industrial, and laboratory environments. It presents the objectives, project overview, benefits, and scope of the project.

Chapter 2 - Literature Review: This chapter reviews existing temperature sensors, gas sensors, and microcontroller-based monitoring systems, discusses the advantages and limitations of traditional and existing methods, and identifies the need for an affordable, compact, and locally operated system.

Chapter 3 - Hardware and Software Requirements: This chapter describes the ESP32 development board, DHT22 temperature sensor, MQ-2 and MQ-5 gas sensors, LCD display, connecting wires, and power supply, along with the Arduino IDE, ESP32 board package, and required libraries.

Chapter 4 - System Design and Methodology: This chapter explains the block diagram, circuit diagram, hardware connections, and working methodology, including sensor initialization, data flow, threshold checking, and LCD updating.

Chapter 5 - Implementation and Results: This chapter presents the hardware assembly, program development, testing of each sensor, recorded observations, and the output of the system.

Chapter 6 - Conclusion and Future Enhancements: This chapter summarizes the achievements of the project and suggests possible improvements such as wireless data transmission, mobile application integration, and cloud-based data storage.

Finally, the report lists the references used, including component datasheets and technical documentation.

2. REVIEW OF LITERATURE

This chapter reviews the technologies and systems related to temperature monitoring, gas sensing, and microcontroller-based environmental monitoring, which form the background of the proposed project.

2.1 TEMPERATURE SENSING TECHNOLOGIES

Temperature sensing is a fundamental part of environmental monitoring. Common electronic temperature sensors include thermistors, thermocouples, resistance temperature detectors (RTDs), analog integrated sensors such as the LM35, and digital sensors such as the DS18B20 and DHT series. Digital sensors are widely preferred in embedded projects because they provide readings in digital form and require little additional circuitry.

The DHT22 is a low-cost digital sensor that measures temperature and relative humidity and communicates with a microcontroller through a single data line. Its simple interface, compact size, and reasonable accuracy make it suitable for indoor environmental monitoring and educational projects.

2.2 GAS SENSING TECHNOLOGIES

Gas sensors detect the presence of particular gases in the surrounding air. Electrochemical, catalytic, infrared, and metal-oxide semiconductor sensors are commonly used for this purpose. Among these, metal-oxide semiconductor sensors of the MQ series are popular in low-cost applications because they are inexpensive, simple to interface, and respond to a range of gases.

The MQ-2 is a general-purpose sensor that responds to gases and vapours such as LPG, methane, hydrogen, alcohol, and smoke. The MQ-5 is mainly sensitive to LPG and natural gas. These sensors contain a small heating element and produce an output that changes with the concentration of the target gases. Their readings are generally used to indicate relative changes in gas levels, and they require a warm-up period and calibration for dependable operation.

2.3 EMBEDDED CONTROLLERS IN MONITORING SYSTEMS

Embedded controllers read sensor outputs, process the data, and control display or communication units. Platforms such as Arduino, Raspberry Pi, and the ESP32 are widely used. The ESP32, developed by Espressif Systems, is a low-cost microcontroller with a dual-core processor, multiple input/output pins, analog-to-digital converters, communication interfaces, and built-in Wi-Fi and Bluetooth connectivity.

These features allow the ESP32 to interface with several sensors at the same time and make it suitable for IoT applications. Even when used for local monitoring, its built-in wireless capability leaves room for future expansion without changing the main hardware.

2.4 TRADITIONAL AND EXISTING MONITORING METHODS

Traditional monitoring methods rely on manual observation using thermometers and standalone gas detectors. These methods require a person to be present, do not combine different measurements in one place, and are not convenient for continuous observation.

Existing electronic monitoring systems improve on this by automating data collection and display. Many of them, however, depend on internet connectivity, cloud platforms, or mobile applications, which increases cost and complexity and makes them unsuitable where a network is unavailable. Commercial instruments are also often expensive and less suitable for learning purposes.

2.5 RESEARCH GAP AND MOTIVATION

The review shows that accurate sensors, capable controllers, and IoT platforms are readily available. However, there remains a need for an affordable, compact, and locally operated monitoring system that combines temperature and gas sensing on a single display without requiring an internet connection. The present project addresses this need by using the ESP32 with the DHT22, MQ-2, and MQ-5 sensors and an LCD display, while keeping the design open for future wireless and cloud-based enhancements.

3. HARDWARE AND SOFTWARE REQUIREMENTS

This chapter describes the hardware and software components required to develop the proposed system and explains the role of each component. The components used are summarized in Table 1.

Table 1: Components used

S.NO	COMPONENT	FUNCTION
1	ESP32 development board	Main controller: reads sensors, processes data, drives the LCD
2	DHT22 sensor	Measures environmental temperature
3	MQ-2 gas sensor	Responds to gases and vapours such as LPG, methane, and smoke
4	MQ-5 gas sensor	Responds mainly to LPG and natural gas
5	LCD display	Displays sensor readings and status
6	Connecting wires	Electrical connections between components
7	Power supply	Provides power to the ESP32 and sensors

3.1 ESP32 DEVELOPMENT BOARD

The ESP32 is a low-cost, low-power microcontroller with a dual-core processor, built-in Wi-Fi and Bluetooth, analog-to-digital converters, and a large number of general-purpose input/output (GPIO) pins. In this project it acts as the brain of the system: it reads the outputs of the sensors, converts and processes the values, compares them with threshold limits, and sends the results to the LCD.

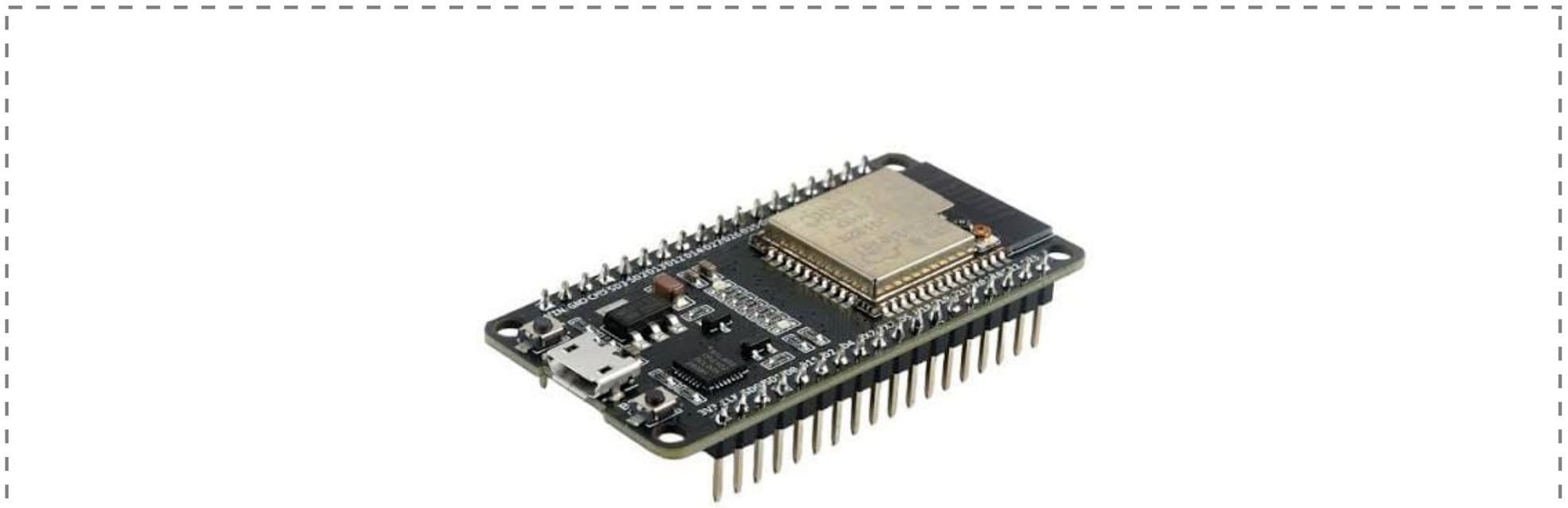


Figure 2: ESP32 development board

3.2 DHT22 TEMPERATURE SENSOR

The DHT22 is a digital sensor used to measure environmental temperature. It contains a sensing element and a small processing chip that outputs the measured value as a digital signal over a single data pin. It is compact, inexpensive, and easy to interface, which makes it suitable for this project.

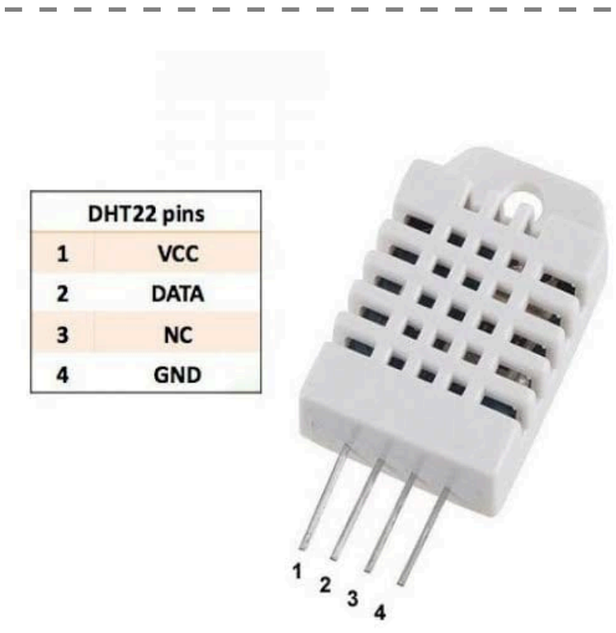


Figure 3: DHT22 temperature sensor

3.3 MQ-2 GAS SENSOR

The MQ-2 is a metal-oxide semiconductor gas sensor that responds to several gases and vapours, including LPG, methane, hydrogen, alcohol, and smoke. A built-in heater warms the sensing element, and the sensor output varies with the gas concentration. The ESP32 reads this output and uses it to assess the air condition.



Figure 4: MQ-2 gas sensor

3.4 MQ-5 GAS SENSOR

The MQ-5 is a similar gas sensor that responds mainly to LPG and natural gas. Using it together with the MQ-2 provides readings from two sensors with different sensitivities, which gives a broader view of the gases present in the surroundings.



Figure 5: MQ-5 gas sensor

3.5 LCD DISPLAY

The LCD display is the output unit of the system. It shows the temperature value, the gas sensor readings, and the environmental status so that the user can observe conditions directly at the monitoring unit without any computer or mobile device.



Figure 6: LCD display

3.6 CONNECTING WIRES AND POWER SUPPLY

Connecting wires establish the electrical connections between the ESP32, the sensors, and the LCD. The power supply provides the required operating voltage to the ESP32 and the sensors; the gas sensors in particular need a steady supply because of their built-in heaters. A stable supply is important for consistent readings.

3.7 SOFTWARE REQUIREMENTS

- **Arduino IDE**: Used to write, compile, and upload the program to the ESP32.
- **ESP32 board package**: Adds support for the ESP32 board to the Arduino IDE.
- **Required libraries**: Libraries for the DHT22 sensor and the LCD display, together with the analog-reading functions used for the MQ-2 and MQ-5 sensors.

- **Program development environment**: Serial Monitor of the Arduino IDE, used to view readings and debug the program during development.

Together, these hardware and software requirements provide the technical foundation for implementing the project.

4. SYSTEM DESIGN AND METHODOLOGY

This chapter explains the design and working methodology of the proposed monitoring system, including the system architecture, circuit connections, and the flow of data from the sensors to the display.

4.1 SYSTEM ARCHITECTURE

The system consists of three main sections: the sensing section, the processing section, and the display section. The DHT22, MQ-2, and MQ-5 sensors form the sensing section and convert environmental conditions into electrical signals. The ESP32 forms the processing section, where the signals are read, processed, and compared with threshold values. The LCD forms the display section and shows the results to the user. A power supply feeds all the sections.The block diagram shows the flow of data from the sensors to the ESP32 and then to the LCD display.

4.2 CIRCUIT DIAGRAM AND HARDWARE CONNECTIONS

The DHT22 provides a digital data signal to a GPIO pin of the ESP32. The MQ-2 and MQ-5 sensors provide analog output signals, which are read through analog-capable input pins of the ESP32. The LCD is connected to the ESP32 for displaying the output. The actual connections used in the project are listed in Table 2.

Figure 8: Circuit diagram of the system

Table 2: Hardware connections between the ESP32 and the components

COMPONENT	COMPONENT PIN	ESP32 PIN
DHT22	VCC / DATA / GND	..... / GPIO .... / GND
MQ-2	VCC / AO / GND	..... / GPIO .... / GND
MQ-5	VCC / AO / GND	..... / GPIO .... / GND
LCD	.....	.....
Power supply	+ / -	.....

4.3 SENSOR INITIALIZATION

When the system is powered on, the ESP32 initializes the DHT22 sensor, configures the analog input pins for the MQ-2 and MQ-5 sensors, and initializes the LCD. The gas sensors require a warm-up period after power-on so that their heating elements reach a stable operating temperature before reliable readings are taken.

4.4 DATA ACQUISITION AND PROCESSING

After initialization, the program enters a continuous monitoring loop. In each cycle the ESP32 reads the temperature from the DHT22 and the analog values from the MQ-2 and MQ-5 sensors. The raw values are processed into a form suitable for display and for comparison with the threshold limits.

4.5 THRESHOLD-BASED STATUS EVALUATION

The program checks the readings against predefined threshold values to determine the environmental status. If the temperature or the gas sensor readings remain within the limits, the status is shown as normal. If any reading exceeds its threshold, the status changes to indicate that the condition requires attention. The threshold values can be adjusted in the program according to the environment and the calibration of the sensors.

4.6 LCD OUTPUT UPDATING

The processed readings and the status are sent to the LCD and refreshed at regular intervals, so the display always shows the latest information. The data flow can be summarized as follows:

- The sensors measure temperature and gas levels.
- The ESP32 reads and processes the sensor outputs.

- The readings are compared with the threshold values.
- The values and the status are displayed on the LCD.
- The cycle repeats continuously.

4.7 OFFLINE OPERATION

The system is designed to perform local monitoring without depending on an internet connection. All measurement, processing, and display functions are carried out by the ESP32 itself, so the system can be used in places where a network is not available.

4.8 OVERALL METHODOLOGY FLOW

System Design → Hardware Connection → Sensor Initialization → Data Acquisition → Data Processing → Threshold Checking → LCD Display → Testing and Evaluation

5. IMPLEMENTATION AND RESULTS

This chapter presents the practical implementation and testing of the project, including hardware assembly, program development, sensor verification, and the observed results.

5.1 HARDWARE ASSEMBLY

The ESP32, DHT22, MQ-2, MQ-5, and LCD were assembled and connected using connecting wires according to the circuit diagram. The connections were checked carefully before the power supply was switched on to avoid incorrect wiring and damage to the components.

5.2 PROGRAM DEVELOPMENT AND UPLOADING

The program was written in the Arduino IDE using the ESP32 board package and the required libraries. It includes sensor initialization, reading of the sensors, threshold checking, and LCD display routines. After compilation, the program was uploaded to the ESP32 development board through the USB connection.

5.3 SENSOR VERIFICATION AND TESTING

The working of each sensor was verified by observing the readings displayed on the LCD. Temperature readings from the DHT22 sensor and gas sensor readings from the MQ-2 and MQ-5

sensors were recorded under different environmental conditions. These observed values were used to evaluate the monitoring performance of the system.

5.4 OBSERVATIONS

The readings recorded under different conditions are given in Table 3.

Table 3: Sensor readings under different environmental conditions

S.NO	CONDITION	TEMP (° C)	MQ-2	MQ-5	STATUS
1	Normal room condition				
2					
3					
4					

5.5 DIFFICULTIES ENCOUNTERED

During implementation, difficulties such as sensor calibration, unstable gas sensor readings during the initial warm-up period, and display issues were encountered. These were addressed by allowing the gas sensors to warm up before taking readings, checking the connections, and adjusting the threshold values and program settings.

5.6 DISCUSSION OF RESULTS

The results show that the system performs the intended monitoring functions. The ESP32 successfully collected the readings from the DHT22, MQ-2, and MQ-5 sensors, processed them, and displayed the results on the LCD. The temperature readings and gas sensor responses changed according to the environmental conditions, demonstrating that the system can be used for basic environmental monitoring.

6. CONCLUSION AND FUTURE ENHANCEMENTS

6.1 CONCLUSION

The IoT Temperature and Gas Monitoring system using the ESP32 was designed and developed to monitor environmental temperature and gas levels and display them locally on an LCD. The ESP32 collects readings from the DHT22, MQ-2, and MQ-5 sensors, processes the data, compares it with threshold values, and displays the environmental information.

The project objectives, including temperature measurement, gas sensor monitoring, local display of readings, and offline operation, were achieved. The system is simple, affordable, and useful for educational and basic environmental monitoring applications. The project also gave practical experience in embedded systems, sensor interfacing, and program development.

6.2 FUTURE ENHANCEMENTS

The system can be improved in several ways to extend its functionality:

- **Wireless Data Transmission:** Sending sensor data wirelessly using the built-in Wi-Fi of the ESP32.
- **Mobile Application Integration:** Monitoring the readings remotely through a mobile application.
- **Cloud-Based Data Storage:** Storing the data on a cloud platform for remote access and analysis.
- **Graphical Data Visualization:** Displaying trends of temperature and gas levels as graphs.
- **Automatic Data Logging:** Recording the readings automatically with time stamps for later analysis.
- **Additional Warning Indicators:** Adding a buzzer or LED indicators to warn the user when thresholds are exceeded.
- **Sensor Calibration:** Improving the accuracy of the gas readings through proper calibration.

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